

Trade Study TS-005

Ground Control Station Telemetry Communication System Selection

Project: Instrument Case Modernization for GCS as a subsystem of TOMTOM

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1. Objective

The objective of this trade study is to evaluate candidate telemetry communication architectures for the Tactical Overwatch Module for Tracking, Observation, and Mapping (TOMTOM) Ground Control Station (GCS) and select the solution that best satisfies the project's operational, technical, and cost requirements. The telemetry subsystem provides the primary communication pathway between the Ground Control Station and the unmanned aircraft, enabling transmission of flight status, vehicle position, navigation information, battery status, mission commands, and other critical system data required for safe and effective mission execution.

Reliable telemetry communications are essential to maintaining situational awareness throughout all phases of operation. The selected communication architecture must provide sufficient range, reliability, data throughput, and compatibility with the Ground Control Station software while supporting future system expansion and integration with mission planning applications.

In addition to satisfying current operational requirements, the selected telemetry architecture should remain compatible with future enhancements including autonomous mission execution, sensor integration, long-range communications, and expanded command and control capabilities while minimizing system complexity and lifecycle cost.

2. Design Requirements

The telemetry communication subsystem shall provide reliable bidirectional communications between the Ground Control Station and the aircraft throughout all mission phases. The architecture must support transmission of MAVLink telemetry, vehicle health information, mission commands, and navigation data while maintaining compatibility with the selected flight control software and Ground Control Station architecture.

Because communication failures directly affect mission safety, reliability was established as the primary design consideration during the evaluation process. The selected architecture should also provide sufficient communication range, low latency, expandability, and ease of integration while remaining compatible with future communication upgrades.

Requirement	Value
MAVLink Compatible	Required
Bidirectional Communication	Required
Portable	Required
Expandable	Required

Low Latency	Required
Long Range Capability	Desired
Linux Compatible	Required
Open Architecture	Required
Budget	<\$300 Preferred

3. Alternatives Considered

Alternative A

SiK Telemetry Radio

SiK telemetry radios are widely used within the open-source UAV community for short-to medium-range communications between Ground Control Stations and autopilots. These radios provide reliable MAVLink communication using license-free frequency bands while requiring minimal configuration and integration effort.

Advantages

- Low acquisition cost
- Excellent MAVLink compatibility
- Large user community
- Simple integration
- Open-source ecosystem

Disadvantages

- Limited communication range
- Lower bandwidth
- Less suitable for future long-range operations

Alternative B

RFD900x Telemetry System

The RFD900x is a professional long-range telemetry radio designed specifically for unmanned aircraft applications. The system supports significantly greater communication distances while maintaining reliable MAVLink communications under demanding operating conditions.

Advantages

- Excellent communication range
- High reliability

- Proven UAV performance
- Expandable
- Strong MAVLink support

Disadvantages

- Higher acquisition cost
- Larger physical size
- Increased power consumption

Alternative C**LTE / 5G Cellular Communications**

Cellular communication systems utilize existing mobile communication infrastructure to provide long-range connectivity wherever network coverage is available.

Advantages

- Extremely long communication range
- High bandwidth
- Internet connectivity
- Supports remote operations

Disadvantages

- Dependent on cellular coverage
- Monthly service costs
- Greater system complexity
- Increased cybersecurity considerations

Alternative D**ExpressLRS Telemetry**

ExpressLRS (ELRS) is an open-source radio communication system originally developed for radio-control aircraft.

Advantages

- Very low latency
- Low acquisition cost
- Excellent open-source support
- Lightweight hardware

Disadvantages

- Limited MAVLink ecosystem maturity
- Additional integration effort
- Less common for Ground Control Station telemetry

4. Evaluation Criteria

The candidate telemetry communication architectures were evaluated using a weighted decision matrix based on communication reliability, system compatibility, implementation complexity, expandability, lifecycle cost, and expected operational performance. Evaluation criteria were selected to ensure the chosen communication system satisfies both the immediate operational requirements of the TOMTOM prototype and future mission capability growth.

Communication reliability received the highest weighting because continuous telemetry is essential for safe vehicle operation and mission execution. Compatibility with MAVLink and the existing Ground Control Station architecture was also assigned significant weighting to minimize software integration effort. Range, expandability, cost, and implementation complexity were evaluated to ensure the selected solution provides the best long-term value while supporting future system evolution.

Criterion	Weight
Reliability	25%
MAVLink Compatibility	20%
Communication Range	20%
Expandability	15%
Cost	10%
Ease of Integration	10%

5. Decision Matrix

Criteria	Weight	SiK	RFD900x	LTE/5G	ExpressLRS
Reliability	0.25	4	5	3	4
MAVLink Compatibility	0.20	5	5	3	3
Communication Range	0.20	3	5	5	4
Expandability	0.15	3	5	4	3
Cost	0.10	5	3	2	5
Ease of Integration	0.10	5	4	3	3

Each alternative was evaluated using a qualitative scoring scale ranging from one to five, where a score of one represents poor performance relative to the evaluation criterion and

a score of five represents excellent performance. Scores were assigned using engineering judgment, published technical specifications, anticipated integration effort, operational performance, and compatibility with the Ground Control Station architecture.

The weighted decision matrix provides a structured and repeatable methodology for comparing candidate communication systems while reducing subjective bias during the engineering decision-making process.

6. Weighted Scores

Alternative	Total Score
SiK Telemetry	4.00
RFD900x	4.70
LTE / 5G	3.50
ExpressLRS	3.80

Weighted scores were calculated by multiplying the score assigned to each evaluation criterion by its corresponding weighting factor and summing the resulting values for each candidate communication architecture.

Although the weighted score serves as the primary decision metric, the final engineering decision also considered hardware maturity, integration risk, future mission expansion, software compatibility, and long-term maintainability.

7. Discussion

The RFD900x telemetry system achieved the highest weighted score among the evaluated alternatives and was selected as the preferred telemetry communication architecture for the TOMTOM Ground Control Station. Although the system has a higher acquisition cost than several competing alternatives, it provides superior communication range, proven reliability, strong MAVLink compatibility, and excellent support within the unmanned aircraft community.

The selected architecture supports robust communication between the Ground Control Station and aircraft while minimizing integration risk and preserving compatibility with future mission software, autonomous operations, and expanded communication requirements. Its modular design also allows future communication technologies to be incorporated without requiring redesign of the overall Ground Control Station architecture.

Lower-cost alternatives such as SiK telemetry provide excellent value for basic UAV applications but offer reduced communication range and scalability. Cellular communication systems provide exceptional range but introduce dependency on

commercial infrastructure and increased operational complexity. ExpressLRS provides excellent latency and affordability but currently offers less mature MAVLink integration than dedicated telemetry systems. Overall, the RFD900x provides the most balanced solution for the TOMTOM Ground Control Station.

8. Decision

Selected Alternative:

RFD900x Telemetry System

Rationale

- Communication reliability
- MAVLink compatibility
- Long-range capability
- Expandability
- Maintainability
- Lifecycle value

The RFD900x telemetry system establishes the communication baseline for the TOMTOM Ground Control Station and provides a reliable, expandable, and mission-capable foundation supporting prototype integration, subsystem verification, and future operational capability development.